

COURSE DESCRIPTION			
Program	Printing Technology		
Course Title	Introduction to Printing Technology		
Course Code	1101	Semester	I
Type of Course	Practicum	Contact Hours	90
Teaching Scheme (L: T:P:R)	3:0:3:0	Credits	4.5
CIE Marks	40	ESE Marks	100

Introduction

This course serves as an introductory platform to Printing Technology, providing students with fundamental knowledge of the printing industry, its processes, machines, and materials through practical exposure, observation, and guided self-learning. The course creates awareness of the role and influence of printing in day-to-day life, including communication, education, packaging, advertising, logistics, and media. It encourages professional curiosity, independent learning, and an understanding of industry standards, practices, and career opportunities

Course Objectives

1	Understand the role and importance of printing in daily life
2	Demonstrate basic printing workflow activities (prepress, press, postpress)
3	Identify basic printing processes, machines, and materials
4	Identify fundamental printing and finishing equipment and its components
5	Follow safe and disciplined laboratory practices
6	Perform basic screen printing operations

Course Pre-requisite

Topic	Course code	Course Title	Semester
NIL			

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Explain the role and evolution of printing in society and industry	Understand
CO2	Demonstrate basic prepress, press and postpress workflow activities	Apply
CO3	Identify printing processes and Practice laboratory safety, discipline, and teamwork.	Apply
CO4	Perform single and multi-colour screen printing with proper registration and safety practices	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	1	-	-	-	1	-	-	-	1	1
CO2	2	2	1	1	1	-	-	-	1	1	-
CO3	2	1	-	-	1	1	1	2	2	-	-
CO4	2	2	2	1	1	-	-	1	1	-	-
Total	8	6	3	2	3	2	1	3	4	2	1
Correlation Level	2	1.5	1.5	1	1	1	1	1.5	1.33	1	1

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	R				CIE	ESE	Total	CIE	ESE	Total	

3	0	3	0	6	90	4.5	20	60	80	20	40	60	140
L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination													

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	P
Module I	Printing Awareness Evolution, & Role in Daily life	Printing - definition and purpose Evolution of printing through the years, Importance of Printing in human civilization. Printing in different sections of Life- Communication, logistics, education, publications, packaging, advertising, culture, office and administration, security documents, Integration of printing with digital media. Printing workflow - Pre press, press and Post press- Introduction	11	0
Module I	Workshop experiments	Printing lab orientation and safety practices. Identification of printed products and their applications	0	2
Module II	Print workflow	Job understanding and planning- Print job requirement analysis, Importance of job planning and documentation Pre press Activities- Content creation and design (basic awareness), Image preparation and layout, Proofing: purpose and types (visual concept only)- Proof reading and proof reading marks, Plate / screen / file (image carrier) preparation (process-level understanding), Equipment used in Pre press Press - machine set up and material testing, machine sections printing and quality checks- Equipment used in press section Post press- Cutting, folding, binding, and finishing, Packaging and dispatch, Importance of post-press in final product quality, Equipment used in post press Plant layout - Concept of printing plant layout- Importance of proper layout in workflow efficiency- Safety, material movement, and productivity considerations	12	0
Module II	Workshop experiments	Demonstrate simple pre-press activities- letter head, Visiting card etc. Draw a detailed sketch of plant layout	0 0	3 3
Module III	Basic Printing Processes	Relief printing: principle, features, applications Planographic printing: principle, features, applications, - Basic concept of offset Intaglio printing: principle, features, applications- Basic concept of Gravure Stencil printing: principle, features, applications Digital printing: basic working concept, Applications Different sections of a Printing machine- Infeed, registration unit, Printing, drying, Delivery (functional role- general View)	11	0
Module III	Workshop experiments	Identification of machine sections. Demonstrate Folding, Cutting operations	0 0	3 3
Module IV	Screen Printing basics	Screen Printing- Definition, advantages, limitations and applications Mesh Materials used- silk, nylon, polyester, metal etc. Frames - wood, metal Screen stretching- manual, machine Stencil - types, preparation methods- Photographic stencils- Substrates used	11	0
Module IV	Workshop experiments	Familiarize with screen printing components , materials and equipment Fabric Stretching on Frame Preparation of hand cut stencils Preparation of Direct stencil. Single color printing practice on screen printing Two color printing practice on screen printing Screen Washing	0 0 0 0 0 0 0	1 6 3 3 3 6 3

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Printing - definition and purpose	Explain the role of printing	L	CO1	Understand	2
Evolution of printing through the years, Importance of Printing in human civilization	Explain History and technological advancement of printing	L	CO1	Understand	3
Printing in different sections of Life	Explain importance of printing in our daily life	L	CO1	Understand	3

Printing workflow	Explain importance of printing in our daily life	L	CO1	Understand	3
Printing lab orientation and safety practices	follow standard laboratory orientation and safety practices by identifying laboratory layout, equipment locations, safety symbols, safe operating procedures applicable to printing workshops	P	CO1	Apply	1
Identification of printed products and their applications	identify different printed products such as books, newspapers, packaging materials, labels, posters, and commercial prints, and explain their applications based on printing process, substrate, and end-use requirements.	p	CO1	Apply	1
Module II					
Job understanding and planning	Create a job card suitable for a printing press	L	CO2	Understand	3
Pre press activities	Explain and differentiate various pre press activities	L	CO2	Understand	2
Press activities	Explain and differentiate various press activities	L	CO2	Understand	2
Post press activities	Explain and differentiate various post press activities	L	CO2	Understand	2
Plant layout	Draw sample plant layout	P	CO2	Understand	3
Demonstrate simple pre press activities- letter head, Visiting card etc	perform basic pre-press activities by designing simple printed materials such as letterheads and visiting cards using appropriate layout principles, fonts, alignment, and page setup suitable for printing	P	CO2	Apply	3
Draw a detailed sketch of plant layout	draw a detailed sketch of a printing plant layout, clearly indicating major departments, machine placement, material flow, and safety pathways as per standard printing industry practices	P	CO2	Apply	3
Module III					
Relief printing	Explain relief principle and its applications	L	CO3	Understand	1
Planography	Explain planography principle and its applications	L	CO3	Understand	2
Intaglio	Explain intaglio principle and its applications	L	CO3	Understand	2
Digital printing	Explain digital printing method and its applications	L	CO3	Understand	2
Screen printing	Explain screen printing processes and its applications	L	CO3	Understand	2
Machine Sections	Identify machine sections	L	CO3	Understand	2
Identification of machine sections	identify major sections of printing and post-press machines and explain the basic function of each section	P	CO3	Apply	3
Demonstrate Folding, Cutting operations	demonstrate basic folding and cutting operations using appropriate machines or hand tools,	P	CO3	Apply	3
Module IV					
Screen printing process	Explain screen printing processes and its advantages	L	CO4	Understand	1
Materials	Explain the materials and tools used in screen printing	L	CO4	Understand	4
Fabric stretching	Explain fabric stretching mechanisms	L	CO4	Understand	3
Stencils	Explain the types of stencils and preparation method	L	CO4	Understand	1
Substrates	Explain screen printing substrates used and its applications	L	CO4	Understand	3
Familiarize with screen printing components , materials and equipment	identify and explain the function of screen printing components, materials, and equipment such as screens, frames, squeegees, inks, substrates, and drying facilities used in manual screen printing	P	CO4	Understand	1
Fabric Stretching on Frame	stretch fabric uniformly on a screen frame, ensuring proper mesh tension and alignment suitable for accurate and defect-free screen printing.	P	CO4	Apply	3
Preparation of hand cut stencils	prepare hand-cut stencils using appropriate stencil materials, accurately transferring simple designs and maintaining clean cut edges for single-color screen printing	P	CO4	Apply	3

Preparation of Direct stencil.	prepare a direct stencil on the screen using suitable stencil emulsions or films, following correct coating, drying, and exposure procedures for image formation.	P	CO4	Apply	3
Single color printing practice on screen printing	perform single-color screen printing(stencil preparation using indirect method) on paper or textile substrates by applying correct ink consistency, squeegee angle, pressure, and stroke to obtain uniform prints	P	CO4	Apply	6
Two color printing practice on screen printing	execute two-color screen printing by registering multiple screens accurately and maintaining color consistency and alignment during the printing process.	P	CO4	Apply	6
Screen Washing	clean and reclaim screens safely by removing ink residues and stencils using appropriate chemicals and washing procedures, ensuring reusability of the screen without mesh damage.	P	CO4	Apply	3

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Collect and classify five printed items	6
2	Reflection: "Why printing is relevant in the digital age"	6
3-4	Write a detailed note on latest technological advancement of printing and its application in our daily life other than just printing on paper	12
Module II		
5	Prepare a flowchart of the printing workflow	6
6	Collect different substrates with different characteristics	6
7-8	Proof read a given Document	12
Module III		
9-10	Prepare a short note on where offset / flexo / gravure printing is used	12
11-12	Prepare a chart showing comparison of different printing processes	12
Module IV		
13	Collection of screen Printed samples	6
14-15	As a project prepare a letterhead for the department	12
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Technology of Offset Printing	C. S. Misra	ggAnupam Prakashan, Prayagraj
2	Printing technology Fifth edition	Micheal J Adams	Delmer

REFERENCE			
Sl. No.	Title	Author	Publication
1	Printing Materials: Science and Technology	Thompson, Kipphan	PIRA International
2	Handbook of Print Media	Helmut Kipphan	Springer

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://print.wye.com/support/solutions/articles/14000157112-basic-stages-of-print-production	II

2	https://www.henghaoprinter.com/blogs/how-does-a-printing-press-work/	I & II
3	https://www.britannica.com/technology/offset-printing	III
4	https://www.britannica.com/technology/flexography	III
5	printwiki.org/Gravure	III
6	https://medium.muz.li/a-beginners-guide-to-screen-printing-by-a-completebeginner-c5bd393d9238	IV

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Single color offset machine	Mini offset	1
2	Screen printing frames		15

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Google doc / Microsoft word/		15

Assessment Methodology - Practicum

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Average of Self learning activities from each module	Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)	Lab Examination (internal)
Duration				3 hours	3 hours	2 hours	2 hours	2.5 hours	3 hours
Exam mark		10	50	40	40	50	50	60	40
Converted to			10	10	10	10	10		
Marks	5	10	10	10		10		60	40
Total	40							100	
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10

Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60